

RESUME

RESUME — HelixCode Session Handoff

Revision: 9 **Created:** 2026-07-08 **Last modified:** 2026-07-28T18:03Z (round-2 review corrections, §11.4.131/§12.10) **Status:** active **Maintainer:** CLI-agent main work stream **Authority:** §11.4.131 Session-resumption file — point a fresh agent here. Composes §11.4.127 / §12.10 / §11.4.65 / §11.4.44.

Revision 9 (2026-07-28T18:03Z) — corrections from an independent round-2 review of bf1ce692 + c2b6945e. That review reproduced this session’s four-quadrant RED/GREEN matrix cell-for-cell and confirmed bf1ce692 clean; it also raised three defects in the rev-8 doc edit, ALL corrected here. **(i) ATTRIBUTION FIXED — rev-8 and rev-9 findings were spliced INTO the Revision 7 banner below, not given their own.** The “REPAIRED in bf1ce692”, bisect-skip, and N2-closure text now sitting inside Revision 7’s paragraph was written in RESUME rev 8 / CONTINUATION rev 20-21, NOT by the Revision 7 / Revision 19 doc-refresh pass — so Revision 7’s own self-description (“this pass touched none of them”) does NOT cover that text. Read it as rev-8-or-later. It is left in place rather than surgically un-spliced (moving it risks losing content); this banner is the correction. **(ii) TIMESTAMP FIXED — Revision 7’s banner stamp was 43 minutes in the FUTURE of its own commit.** dc9e4caa committed at 2026-07-28T17:37:47Z but stamped itself 18:20Z, which made rev 8 (17:45Z) read as “modified before” rev 7. The Revision 7 stamp below is corrected to its true commit time. Forward-fix only — no history rewrite (§11.4.113). **(iii) ANCHORS REFRESHED — Revision 7’s snapshot numbers are STALE and MUST NOT be quoted.** Live as of this revision: meta-repo HEAD **683205eb**, 25 commits ahead of all remotes (Revision 7 says 99ff7d8e/20 — superseded). helix_agent **5b88aea6**, **FULLY PUSHED (0 ahead)** — its 13-commit delta reached a CLEAN GO on review round 3 and is published; the meta pin was bumped to match in 683205eb. debate_orchestrator **2a9485e**, **2 ahead, still UNPUSHED** — round-2 review returned Ready-to-merge=Yes with one optional Minor, which is being closed before push; its pin is deliberately NOT bumped while unpushed (a pin to an unpushed commit dangles for any clone). tool_schema 8cec90b, pushed. **Re-derive these rather than quoting them** — four subagent streams were committing concurrently as this was written, so the numbers drift by design. **(iv) STILL OPEN, carried forward honestly:** an independent review found the replica-RED bluff

class STILL LIVE in two sibling files (internal/server/
llm_default_model_regression_test.go:103,127,170 —
oldResolveDefaultModelReplica reduces to "" == ""; and
llm_generate_regression_test.go:187-198). Conversion is in
flight. Note the discipline point: 3fd55a4d's claim that its own bluff was
a "tracked follow-up" was verified UNBACKED — zero matching items
in docs/workable_items.db or docs/Issues.md (§11.4.148). Prose is
not tracking.

**Revision 7 (2026-07-28T17:37Z) is a CORRECTION revision produced
by a dedicated doc-refresh pass** (scope: this file + docs/
CONTINUATION.md only, dispatched while four OTHER streams actively
worked helix_code/internal/llm/, helix_code/applications/,
submodules/helix_agent/internal/agents/, and submodules/
debate_orchestrator — this pass touched none of them, read-only
against the rest of the tree). Revision 6's numbers are STALE — the repo
kept moving hard after it was written. Independently re-derived this
pass (§11.4.6, nothing copied forward): **(A) Meta-repo HEAD moved
from c29e1dcc to 99ff7d8e — 12 more commits, main is now 20
commits ahead of all 4 remotes**, not 8. New commits include a self-
inflicted-and-fixed §11.4.108 SOURCE → ARTIFACT gap: 3fd55a4d
recovered an orphaned regression guard whose composite literal
referenced a struct field (respModel) that did not exist in the
committed tree — internal/server did NOT compile at 3fd55a4d/
3c8197cf/905a0b0a — caught by independent review and fixed one
commit later in 34e264e1. 3fd55a4d's own commit message
additionally DISCLOSES that its RED_MODE=1 branch is itself a bluff
(asserts on a local in-test replica of the pre-fix response construction
rather than driving the real handler, so it passes on already-fixed code)
— **REPAIRED in bf1ce692**: both polarities now drive the REAL handler
over one shared fixture, proven by the full four-quadrant matrix (pre-
fix RED=PASS defect-reproduced / pre-fix GREEN=FAIL / fixed
RED=FAIL / fixed GREEN=PASS), with the pre-fix artifact reconstructed
in a throwaway git worktree with submodules/ symlinked in. An
independent review also found the "tracked" claim in 3fd55a4d's
message was UNBACKED — no entry existed in docs/, Issues.md, or
the workable-items DB (§11.4.148); it was fixed rather than back-filled.
Bisecting across 3fd55a4d..905a0b0a needs git bisect skip —
those three commits do not build internal/server, so a bisect run
there reports a BUILD failure, not a test verdict, and must not be read
as the defect's location. c7484cc7 (scaling-harness signal-to-noise,
Trials 3 → 5, confirmed in source: MeasureTrialsPerStep = 5) and
640db264 (aurora_os uint64-underflow fix) both carry pasted
verification output in their own commit messages and are treated as

independently-clean per the dispatching brief; this pass did not re-run an independent review of them. The newest commit, 99ff7d8e, is a review artefact in its own right: reviewing 3c8197cf's new *_racefix_test.go files by READING (this host cannot build Fyne/glfw — no X11 dev headers) found the harmony_os test panics deterministically on first execution (nil apiClient dereference in refreshData()) and fixed it — proof that the “authored but never executed” GUI test gap (below) can hide real defects, not just theoretical ones. **(B) helix_agent moved from a345c551 to f451d342 — 9 more commits, now 11 commits ahead** (was 2), working tree still dirty (18 porcelain entries, not re-itemised this pass — re-derive). The dispatching brief for this task described a Critical review finding as “remediation in flight” (Dreamer's memoryMu not covering cleanupPhase's unlocked MEMORY.md read-trim-write); **this pass verified by diff that the fix has actually LANDED**, in f451d342 — cleanupPhase now goes through the same beginMemoryWrite()/end() helper as saveMemories(), with a dedicated memoryWritersPeak runtime-signature test proving no concurrent writers. What this pass could NOT verify from git alone: whether f451d342 and 129094b0 (the 2 commits landed after the reported 9-commit/3-round CLEAN GO at 0367570e) have themselves been through independent review — no review artefact was found. Treat that as open, not assumed clean. **(C) tool_schema is now FULLY PUSHED** — 8cec90b, 0 commits ahead (was 1), clean, confirmed via ls-remote on all 4 configured remotes. **(D) debate_orchestrator is tracked in this document for the first time.** b6c90e7, 0 ahead, clean, confirmed pushed via ls-remote — but it carries only ONE remote (origin → github.com:HelixDevelopment/DebateOrchestrator.git), unlike every sibling submodule's 4-remote (github/gitlab/origin/upstream) fan-out. Flagging this for the operator as a decision point, not fixing it unilaterally — no scope in this pass to add remotes to a submodule four other streams may be relying on. **(E) The scaling-harness stability gap the dispatching brief described (no -count>=10 sweep at the shipped Trials=5, one run reportedly hung at ~0.3% CPU and killed without diagnosis) could NOT be independently corroborated this pass** — no matching log, process, or docs/qa artefact was found for the “hung at 0.3%” detail specifically. It is carried forward as reported, unconfirmed by this pass's own evidence, and should be re-verified rather than trusted at face value (§11.4.6) — the underlying gap (no -count>=10 stability run on record) IS independently confirmed: no such sweep log exists under qa-results/. **(F) Hard-won lesson from this session, worth stating plainly for whoever reads this next: a RED test that asserts against a local replica of the old behaviour, rather than driving the real**

production code path, is a bluff gate — it can never fail, so it proves nothing. This surfaced independently at least twice this session in slightly different shapes: 3fd55a4d's own disclosure above, and (per the dispatching brief, not independently re-verified by this pass) an equivalent pattern in `debate_orchestrator's protocol_convenience_form_red_test.go`, repaired in that submodule's b0655d3. When authoring or reviewing a §11.4.115 RED/GREEN polarity test, confirm the RED branch calls into the actual handler/function under test — not a hand-rolled reconstruction of what it used to do. Everything else in Revision 6 below not superseded by (A)-(F) above is unchanged and still holds — the constitution-gate-sweep GREEN result, the G7 resolution, the platform-up state, and the corrections it made to Revision 5 were all reconfirmed live during this pass (see the updated Live-state anchors table) and are not repeated here.

Revision 6 (2026-07-28T12:35Z) is a CORRECTION revision from a read-only audit. Revision 5's central claims were ALREADY STALE by the time they were read — the repo was (and, as of this revision, still is) being actively advanced by other concurrent streams/agents while revision 5 was being written and after. Verified corrections: **(A) G7 NOW PASSES — 0 violations, not 13.** Independently re-run (`bash scripts/verify_qa_evidence.sh --enforce --since 925169c98945ca0fee1e84dae53ad494e4897832`): 28 feature-shipping commits evaluated, **0 violations**, 0 opt-outs. The retrospective QA-evidence commits rev 5 called “Class A/B/C, still violating” (including all 3 security fixes) are landed and now report ok. **The full constitution gate sweep is GREEN:** `bash scripts/verify-all-constitution-rules.sh` → 16 gates run, **15 PASS + 1 honest SKIP** (G14, docs_chain engine absent, SKIP-OK per §11.4.3) — **zero FAIL**, exit code 0. **(B) HEAD IS NOT 0a4eb8d0 AND NOT EVERYTHING IS PUSHED.** Rev 5's “nothing unpushed” was true only at 0a4eb8d0. The main-stream track kept committing after rev 5 was written; as of **2026-07-28T12:34:54Z**, HEAD is c29e1dcc and **local main is 8 commits ahead of every configured remote** (github/gitlab/origin/upstream — all 0 behind). This count is VOLATILE — HEAD moved 3 times (0a4eb8d0 → e952f4d1 → a5462e35 → c29e1dcc) during this single audit; re-derive with `git log --oneline origin/main..HEAD` before trusting it. **(C) Constitution pin now MATCHES the checkout** (ce3331a1, both), not “6 behind” — resolved by commit 925aa859. **(D) The §11.4.157 carrier-lockstep gap is effectively closed.** 5 of 6 root carriers (CLAUDE.md/AGENTS.md/QWEN.md/GEMINI.md/CONSTITUTION.md) are committed at HEAD reaching §11.4.235 (matching the constitution submodule's own ceiling); CRUSH.md is

committed at §11.4.234 but its working-tree edit (already present, uncommitted) already reaches §11.4.235 — one commit away from full lockstep. **Everything else in rev 5 not superseded above (Traps, Sweep commands, Corrections 1-5, Binding constraints) is unchanged and still holds.**

SHORT variant (one paste, first sentence)

Read RESUME.md then docs/CONTINUATION.md, run `git fetch --all --prune --tags`, and continue on main (Track 1 — §11.4.182 TRUNK RULE: trunk work is ALWAYS (T1/main - <alias>)). HEAD is 99ff7d8e as of 2026-07-28T17:37Z (re-derived this pass — this repo has moved fast every time it has been checked; re-derive again with `git rev-parse HEAD` before trusting this number). **main is 20 commits ahead** of every configured remote (github/gitlab/origin/upstream, all 0 behind) — re-derive with `git log --oneline origin/main..HEAD`. `helix_agent` is **11 commits ahead** of its 4 remotes (dirty working tree, 18 porcelain entries); `tool_schema` is now **fully pushed** (0 ahead, was 1); `debate_orchestrator` is **fully pushed** but has only ONE remote (github), unlike its 4-remote siblings — flag for the operator, don't fix unilaterally. The working tree carries **73 porcelain entries** (46 untracked + 27 modified — mostly in-flight docs/qa/<run-id>/evidence dirs from a concurrent stream's G7 work; account for every one before `git add`, NEVER `git add -A`, §11.4.84). **The platform is UP and systemd-ENABLED** — 5 units active, ports 7061/8081/8100/8443 bound (reconfirmed this pass, unchanged); do NOT boot infra blindly, check first. **What blocks the tag:** (1) the §11.4.40 full Go-test retest on a QUIET host has still not concluded (the durable `helix_code_inner` sweep log is unchanged since 14:41 — hours before the newest commits landed; a fresh serial run is needed, not a re-read of that log); (2) pushing the now-20-ahead main (+ `helix_agent`'s 11-ahead) before choosing a tag point; (3) §11.4.185 manual QA-team confirmation, for which **no record exists in this repo**; (4) a disclosed-but-unrepaired RED bluff in 3fd55a4d's F1 guard (asserts on a local replica, cannot fail) still needs its RED branch rewritten to drive the real handler; (5) three `main_racefix_test.go` files (desktop/harmony_os/aurora_os) remain authored-but-never-executed on this host (no X11 dev headers) — one was proven to hide a real bug (fixed in 99ff7d8e) by reading alone, so treat the other two's coverage as unconfirmed. Force-push is absolutely forbidden (§11.4.113).

FULL variant (paste-ready block)

You are resuming the HelixCode test-suite remediation cycle on main = **Track 1** (§11.4.182 TRUNK RULE, operator mandate 2026-07-28: trunk work labels as (T1/main - <alias>), never (T?/...)). The prior sessions landed real fixes with captured RED → GREEN evidence, and also made several attribution errors that are corrected below (and in Revision 6/7's blockquotes above) — read the corrections before trusting any earlier summary.

Read first: RESUME.md (this file), docs/CONTINUATION.md. Then `git fetch --all --prune --tags` and `git log --oneline HEAD..@{u}` (expect empty — the risk is the reverse direction, @{u}..HEAD, see below).

State (as of 2026-07-28T17:37Z, doc-refresh pass — RE-DERIVE, do not trust, this repo moves fast): meta-repo main @ 99ff7d8e, **20 commits ahead** of all 4 configured remotes (github/gitlab/origin/upstream — `git log --oneline origin/main..HEAD`); helix_agent @ f451d342 (**11 commits ahead** of all 4 of its remotes; working tree DIRTY, 18 porcelain entries, not re-itemised this pass); tool_schema @ 8cec90b (**fully pushed, 0 ahead**, clean — was 1 ahead in rev 6); debate_orchestrator @ b6c90e7 (**fully pushed, 0 ahead**, clean, but only ONE remote configured — github — unlike siblings' 4-remote fan-out; flag for operator, newly tracked in this document); helix_qa @ 88ef0579 and challenges @ 072724af (both clean, 0 ahead/0 behind on all remotes, reconfirmed unchanged); constitution checkout @ ce3331a1 — pin still MATCHES the checkout (unchanged, reconfirmed). The meta-repo working tree carries **73 porcelain entries** (46 untracked + 27 modified; re-derive with `git status --porcelain`) — overwhelmingly new docs/qa/<run-id>/ evidence directories from a concurrent stream's ongoing G7 work, plus a handful of source edits from other active streams (per this task's scope fence: helix_code/internal/llm/, helix_code/applications/, and other in-flight edits belong to OTHER streams — do not stage them).

Terminal goal: tag helix-code-1.2.0-dev-0.0.1 on the main repo AND every owned submodule carrying changes, identical helix-code-prefix (§11.4.151), fast-forward-only to all upstreams (§11.4.113) — **conditional on the sweep being green**, reaffirmed by the operator: **tag ONLY when genuinely green**. The constitution-gate-sweep component was GREEN as of rev 6's audit (16 gates, 15 PASS + 1 honest SKIP, 0 FAIL) and was not re-run this pass (out of scope — doc-only). **The tag is still NOT authorised to proceed.** Beyond rev 6's blockers (a full Go-test retest of helix_code + helix_agent on a QUIET host — the durable

helix_code_inner sweep log is unchanged since 14:41, hours before the newest commits landed, so it does NOT cover current HEAD; §11.4.185 manual QA-team confirmation, still no record in this repository; the unpushed-commit counts, now 20/11/0/0 for main/helix_agent/tool_schema/debate_orchestrator), this pass surfaces two ADDITIONAL open items: a disclosed RED bluff in 3fd55a4d (the F1 guard’s RED_MODE=1 branch asserts on a local replica and cannot fail — tracked, not yet repaired) and unconfirmed independent review of helix_agent’s newest 2 commits (129094b0, f451d342) landed after its reported 9-commit/3-round CLEAN GO. Also still flagged for the operator: the target version 1.2.0-dev-0.0.1 is a MINOR bump past the existing 1.1.0-dev-0.0.1/.2/.3 sequence — no CHANGELOG entry or other in-repo rationale for the minor bump (vs. continuing 1.1.0-dev-0.0.4) was found; confirm this is intended.

Anti-bluff is binding (§11.4): every closure needs pasted runtime output; every gate touched needs a paired §1.1 mutation; no assertion may be weakened and no bare `t.Skip()` added to force green. **This session’s hard-won lesson: a RED test asserting against a local replica of old behaviour rather than driving the real code path is a bluff gate that can never fail** — check this specifically when reviewing any §11.4.115 polarity test.

Live-state anchors

All values below verified against the live system on **2026-07-28T12:34:54Z** by a read-only audit. Re-derive before trusting (§11.4.6) — this table has been wrong before, AND the repo was actively moving DURING this very audit (HEAD advanced 3 times in ~10 minutes of observation).

Key	Value	How verified
Meta-repo HEAD	rev 6 row superseded — re-derived rev 7 (2026-07-28T17:37Z, doc-refresh pass): 99ff7d8e (branch main) — 20 commits ahead of ALL 4 configured remotes (github/gitlab/origin/upstream), 0 behind (was c29e1dcc/8-ahead at rev 6, 12 more commits landed since — see the Revision 7 blockquote at the top of this file for what they contain, incl. a self-inflicted-and-fixed compile break at 3fd55a4d → 34e264e1). VOLATILE — this repo has moved every single time it has been checked this session; re-derive again	git rev-parse HEAD git prune --tags then git right --count main main per remote

Key	Value	How verified
Working tree	superseded — re-derived rev 7: 73 porcelain entries (46 untracked + 27 modified; was ~68 at rev 6) — overwhelmingly new docs/qa/<run-id>/ untracked evidence dirs from a concurrent stream’s ongoing G7 work, plus source edits belonging to OTHER active streams per this task’s own scope fence — VOLATILE, re-derive, never quote back, and never git add -A	git status --porcelain
helix_agent	superseded — re-derived rev 7: f451d342 — 11 commits ahead of all 4 of its remotes (was a345c551/2-ahead at rev 6; 9 more commits landed, including the reported CLEAN-GO range ending 0367570e plus 2 further commits 129094b0/f451d342 whose independent review this pass could NOT confirm from any git-visible artefact); working tree DIRTY (18 porcelain entries this pass, not re-itemised — re-derive fresh rather than trusting rev 6’s “14 modified + 4 untracked” breakdown). CONFIRMED THIS PASS by reading the diff: the Critical review finding described elsewhere as “remediation in flight” (Dreamer memoryMu not covering cleanupPhase’s unlocked MEMORY.md write) has actually LANDED, in f451d342 — cleanupPhase now goes through the same beginMemoryWrite()/end() helper as saveMemories()	git -C submodule log/fetch; git -C show --stat f451d342
helix_qa	88ef0579 (gitlink matches worktree); clean; 0 ahead/0 behind — reconfirmed rev 7, unchanged	git -C submodule
challenges	072724af (gitlink matches worktree); clean; 0 ahead/0 behind — reconfirmed rev 7, unchanged	git -C submodule fetch
tool_schema	superseded — re-derived rev 7: 8cec90b — fully pushed, 0 commits ahead (was 1-ahead at rev 6) of all 4 remotes (vasic-digital), confirmed via ls-remote; clean working tree	git -C submodule log/fetch; git -C ls-remote --exit-code heads/main
debate_orchestrator		

Key	Value	How verified
	NEW this revision — not tracked in rev 6 at all. b6c90e7 — fully pushed, 0 ahead, clean, confirmed via ls-remote. Carries only ONE configured remote (origin → github.com:HelixDevelopment/DebateOrchestrator.git), unlike every sibling submodule's 4-remote (github/gitlab/origin/upstream) fan-out — flagged for the operator as a decision point, not changed unilaterally in this pass (out of scope: this task's scope fence explicitly excludes submodules/debate_orchestrator)	git -C submodule status/log/fetch debate_orchestra
constitution — checked out	ce3331a1 (v1.0.0-18-gce3331a); in sync with all 8 remote-tracking refs (ahead=0 behind=0, fresh fetch this audit)	git -C constitution remote rev-list
constitution — meta-repo pin	ce3331a1 — MATCHES the checkout exactly. RESOLVED (was 731bf1d3, 6 behind, in rev 5 — fixed by commit 925aa859). The only remaining dirt under constitution is its OWN nested submodule submodules/session_orchestrator (checked out at 6961c998, detached, but constitution's own index pins 78b77bb8 — a minor, unrelated nested-submodule drift, not a helix_code release blocker)	git ls-tree HEAD constitution rev SHAs)
Constitution ceiling vs carriers	constitution defines §11.4.235; 5 of 6 root carriers (CLAUDE.md, AGENTS.md, QWEN.md, GEMINI.md, CONSTITUTION.md) are COMMITTED at HEAD reaching §11.4.235 already — RESOLVED for those 5. CRUSH.md is committed at §11.4.234 but its working-tree edit (present, uncommitted) already reaches §11.4.235 — one commit away from full lockstep, not an open gap requiring new work	git show HEAD:< '11\4\.[0-9]+' tail -1 per carrier tree copy
Release prefix	helix-code (from HELIX_RELEASE_PREFIX in .env, mode 0600 — confirmed unchanged)	grep '^HELIX_RELEASE_PREFIX'
Latest existing tag	helix-code-1.1.0-dev-0.0.3 → stated target helix-code-1.2.0-dev-0.0.1 is prefix-consistent but is a MINOR bump past the existing -1.1.0-dev-0.0.{1,2,3} sequence; no CHANGELOG/doc rationale	git tag -l 'helix-code-1.2.0-dev-0.0.1' (no hits)

Key	Value	How verified
	found for skipping -1.1.0-dev-0.0.4 — flag for explicit operator confirmation	
Platform	UP and ENABLED — unchanged from rev 5, re-verified this audit (helixagent/helixcode-server/helixllm-gateway/helixcode-infra/helixllm-coder enabled+active; helix.target enabled+inactive; ports 7061/8081/8100/8443 bound)	systemctl --usep -ltnp
Constitution gates	16 run, 15 PASS + 1 honest SKIP (G14) — ZERO FAIL as of rev 6's audit (2026-07-28T12:34Z). NOT re-run this pass (doc-refresh scope only, per this task's brief) — treat as last-known-good, not reconfirmed at current HEAD 99ff7d8e	rev 6: bash script constitution-run scripts/verify_c --since 925169c98945ca0
Sweep logs (current)	Still true as of rev 7 — re-checked, unchanged. qa-results/full_retest/newest entries as of this pass are all_captures_20260728T104042Z.log (15:41) and the same helix_code_inner_20260728T093852Z.log (mtime still 14:41) noted at rev 6 — hours before the newest commits (3fd55a4d through 99ff7d8e, timestamped 21:59-22:18) landed. This durable log does NOT cover current HEAD; a fresh serial go test run is required, not a re-read of it. Whether PID 705363's wait-wrapper is still polling was not re-checked this pass (ephemeral, unlikely to still be the same PID hours later)	ls -lat qa-resul
Manual QA-team confirmation (§11.4.185)	NO RECORD FOUND anywhere in this repo (docs/qa/, RESUME.md, CONTINUATION.md) of an actual manual QA-team sign-off for this release cycle. UNKNOWN whether it happened out-of-band; settle by asking the operator or checking any external QA tracker	grep -rliE "manu sign.?off\ appro
Pre-gofmt tree backup	<scratchpad>/helix_agent_pre_gofmt.patch (3322 lines) — path not re-verified this revision either	—

Operator decisions recorded 2026-07-28 (binding, newest first)

1. **helixcode-infra becomes the SOLE infra owner** — resolves the competing-orchestrator blocker (task #9). Set `HELIX_AUTOBOOT_INFRA=false` so `cmd/server` does not race a second boot (the flag is honoured at `helix_code/internal/infraboot/infraboot.go:118-125`), and `helix_code/config/replica-8081.yaml` is corrected (uncommitted, +102/-5).
2. **G7 is cleared by REAL retrospective QA runs — the 3 security commits first.** Not a baseline bump. See the G7 section.
3. **helixcode-server secret env names are reconciled.**
4. **Tag ONLY when genuinely green.** Reaffirms the conditional authorisation; the sweep is not green, so do not tag.

Operator mandate 2026-07-28 — Track identity. Work on main is **ALWAYS Track 1**; the label is (`T1/main - <alias>`), never (`T?/main - ...`), regardless of checkout path. Landed in `constitution/Constitution.md §11.4.182` as the TRUNK RULE and pushed; the constitution checkout is in sync with all 8 upstreams (`ahead=0/behind=0` on every remote-tracking ref, as of last fetch — no fetch was performed while writing this revision).

Operator decisions recorded 2026-07-27 (binding)

1. **Ports** — do NOT hand-assign. Use the **containers submodule's dynamic-port + service-discovery** capability; extend that submodule for anything missing (§11.4.74 extend-don't-reimplement) and cover the extension with all supported test types. All systems/services/infrastructure choose an available port at runtime, bind it, and expose it through discovery; services discover each other. This supersedes the earlier “which service moves off 8100” framing.
2. **Constitution pin** — advance to the checked-out HEAD in this release.
3. **Moving baselines** — fix the tests so they stop writing to tracked files (root-cause fix, not gitignore).
4. **Compose coupling** — relocate **mcp_servers** under `helix_agent` so no upward path traversal is needed.

Landed this session, with captured evidence

- **gofmt across our Go code:** 507 → 0 unformatted. Vendored/archival trees deliberately EXCLUDED and reverted (see Correction 2). 431 format-only files.

- **Production nil-deref crash FIXED** — internal/llm/llamacpp_provider.go GenerateStream.req, _ := http.NewRequestWithContext(...) discarded the error, so the next line dereferenced a nil *http.Request. Fault address 0x38 == offsetof(http.Request.Header), confirming it exactly. Guarded by internal/llm/llamacpp_stream_malformed_url_test.go (§11.4.115 RED_MODE polarity test). **Proven:** RED_MODE=1 PASS pre-fix / FAIL post-fix; RED_MODE=0 FAIL pre-fix / PASS post-fix; whole package ok 69.789s.
- **G8 gate reconciled** (§11.4.120) — was a FALSE POSITIVE, see Correction 3. Now PASS, with a 7-assertion paired mutation.
- **G13 now 82/82** — four honest Sources-verified footers, zero invented URLs, two recording explicit non-verification gaps.

Platform state — VERIFIED 2026-07-28 (supersedes the 2026-07-27 “shutdown” claim)

THE PLATFORM IS UP AND systemd-ENABLED. Revision 4 of this file claimed “*PLATFORM IS DOWN AND DISABLED ... nothing auto-starts*”. That is **FALSE** as of **2026-07-28**. Live readings:

Unit	enabled	active
helixagent.service	enabled	active (running) — HelixAgent on :7061
helixcode-server.service	enabled	active (running) — HelixCode server on :8081
helixllm-gateway.service	enabled	active (running) — LLM router
helixcode-infra.service	enabled	active (exited) — oneshot, infra up
helixllm-coder.service	enabled	active (exited) — oneshot
helix.target	enabled	inactive
helix-{pg,redis}-data-volume.service	generated	inactive

Bound ports (ss -ltnp): **7061** (helixagent), **8081** (helixcode), **8100** (llm-verifier), **8443** (helixllm), plus :8080 with no owning user process visible.

Consequence for a resuming agent: do NOT blindly boot infra — it is already up, and a second orchestrator racing the running one is exactly the task-#9 blocker below. Check `systemctl --user is-active` and `ss -ltnp` FIRST. Equally, do not assume the ports named in the older record are free: 8100 is held by `llm-verifier`, not by `HelixAgent`.

Untouched by design (other projects on the shared host, §11.4.174): `helixterm-*` + `77c45f5c104d-infra`, `penpot-*`, `helix_sonarqube*`, and **`atmosphere-aosp-build`** — do not assume the host is idle, and never kill a process you have not proven is ours (§11.4.174).

Session 2026-07-27 evening — additions and corrections

Landed — PORT_CONFLICT ownership fix (RED → GREEN captured)

`tests/precondition/containers_boot_test.go` + new `tests/precondition/port_conflict_ownership_test.go`. The gate held **two opposite §11.4.201 defects**:

- **Fail-closed:** probed `{port, port+10000}` and declared `PORT_CONFLICT` if both answered, with zero ownership check. Our stack publishes postgres on `15432 = 5432+10000`, so *any* foreign postgres on the standard port (here `helixterm`) produced a false failure.
- **Fail-open:** `findContainerInstances` ran `docker compose ps --filter name=...` (because `detectContainerRuntime` returns `args ["compose"]`) — not a container-listing command. It returned nothing, so the duplicate-instance check **silently never fired**.

Replaced with runtime-driven ownership: `listRunningContainers` → `ourServiceInstances` (filters the `helixagent-` prefix) → `portConflict`. Evidence: `RED_MODE=1` reproduces (“*legacy blind probe flags a foreign listener as our own duplicate*”), `RED_MODE=0` GREEN, 3/3 both polarities, ok `dev.helix.agent/tests/precondition 0.005s`. The real-duplicate case is still caught (§11.4.120 — not weakened to a tautology).

Correction 4 — gate status in the 1334Z log is STALE

Sweep `qa-results/full_retest/verify_rules_20260727T184002Z.log` (16 gates, 1 failure): **G1 PASS, G8 PASS, G11 PASS, G12 PASS, G13 PASS (82/82 footered)**, G15/G16 PASS, **G14 SKIP** (docs_chain engine absent — SKIP-OK). Only G7 fails — at 18 violations in that log, and at 13 in the fresher `/tmp/g7-qa.out`

(2026-07-28 14:07). Re-read the log, and re-run the sweep to regenerate the volatile detail file, before acting on any earlier gate claim.

Correction 5 — the discovery “failures” were never code defects

internal/discovery TestAllocatePort_RangeExhausted_WithEphemeral / TestConcurrentAllocations / TestConcurrentReleases pass on a quiet host: ok dev.helix.code/internal/discovery 0.013s. The 1332Z run launched the helix_code (133220Z) and helix_agent (133218Z) sweeps **2 seconds apart**; helix_agent’s stress/chaos saturated the ephemeral range 32768-60999, so the tests’ preferred ports 50000-50049 were all unavailable (forcing fallback into the 19-port api range → exactly the 16 unique ports logged) and net.Listen(":0") returned EADDRINUSE. Independent confirmation of Correction 1.

NEW blocker — competing container orchestrators (task #9, architectural)

helixagent.service’s container adapter recreates helixagent-postgres from **docker-compose.yml**, where postgres has **NO ports: section** — so 15432 is never published. Manual docker-compose.test.yml runs DO publish it. With both in play postgres crash-loops: could not bind IPv4 address "0.0.0.0": Address in use (host 5432 is held by the unrelated helixterm stack). **While the service runs, the precondition gate cannot see 15432.** Compounding it, scripts/ensure-infrastructure.sh health-checks \${DB_PORT:-15432} / \${REDIS_PORT:-16379} but boots the default compose — it cannot satisfy its own health check (task #7).

RESOLVED by operator decision, 2026-07-28 — no longer awaiting a call.

helixcode-infra becomes the **sole infra owner**: set

HELIX_AUTOBOOT_INFRA=false so cmd/server does not race a second boot (honoured at helix_code/internal/infraboot/infraboot.go:118-125 — returns Skipped: true, Reason: "disabled via HELIX_AUTOBOOT_INFRA"), and helix_code/config/replica-8081.yaml is corrected (edit present in the working tree, +102/-5, uncommitted). Remaining work is to land and verify that configuration, not to decide it.

Determinism run 2026-07-28 — the 5 “failing” packages were CONTENTION, not defects

qa-results/full_retest/determinism_20260728T093338Z.log — each package run 3× **on a quiet host**, every run exit=0:

Package	Runs	Result
internal/tools/shell	3/3	ok (3.577s / 3.561s / 3.563s)
internal/verifier	3/3	ok (4.305s / 4.307s / 4.307s)
internal/worker	3/3	ok (43.419s / 43.385s / 43.384s)
tests/scaling	3/3	ok (5.677s / 5.700s / 5.636s)
tests/stresschaos	3/3	ok (0.936s / 0.938s / 0.944s)

15/15 PASS. These are **contention artefacts, not code defects** — independent confirmation of Correction 1 (§11.4.119). Do not open defect items against them; do re-run serially before believing any future red from this set.

Sweep results — status by stream

Stream	Verdict	Detail	As of
helix_qa	GREEN (not re-verified this audit)	142 ok / 0 FAIL / 23 no-test-files; submodule itself clean + in sync on all remotes	2026-07-27
governance cascade	GREEN (not re-run this audit)	277 PASS / 0 FAIL; 227 anchors discovered; carriers NOW at §11.4.235 (5/6 committed, CRUSH.md pending commit — see Live-state anchors), superseding “all 6 to §11.4.234”	2026-07-27
constitution rules	GREEN — audit re-run, exit 0	16 gates, 15 PASS + 1 honest SKIP (G14), ZERO FAIL. G7 independently re-verified 0 violations (28 commits evaluated). SUPERSEDES the 2026-07-27 23:40 log below	2026-07-28T (this audit)
helix_code inner	UNKNOWN / IN-FLIGHT — not concluded by the owning stream as of this audit	Newest log unchanged since 14:41 (helix_code_inner_20260728T093852Z.log); a companion wait-wrapper (PID 705363) has been polling since 14:21 without concluding (see Live-state anchors — likely a self-match pgrep bug, not a still-running test). This audit does NOT compute or claim a pass/fail tally for this stream — that determination belongs to the stream that dispatched it	2026-07-28, not concluded
helix_agent	UNKNOWN / AMBIENT-SUSPECT — not a quiet-host run	A go test -count=1 -timeout=900s ./... run (PID 2226428, 17:23 → ~17:39) against a DIRTY working tree completed with 360 ok / 18 FAIL packages, including tests/integration hitting the exact 900s timeout wall. The host was	2026-07-28T contended h

Stream	Verdict	Detail	As of
		NOT quiet during this run (this audit's own fetches + other concurrent activity were in progress). Per this file's own "Correction 1" precedent, this pattern (timeout-exact-hit + assorted package FAILs on a contended host) is ambient-suspect, NOT a confirmed defect list — a serial quiet-host re-run is required before treating any of these 18 as real	

Net effect of this audit: the constitution-gate-sweep component of “the full sweep” is now fully GREEN (G7 included). What remains open is the Go-test component (§11.4.40) on a genuinely quiet host, which neither this audit nor the two concurrently-running streams had concluded as of 2026-07-28T12:34:54Z.

CORRECTIONS — read these before trusting earlier summaries

Correction 1 — three concurrent sweeps contaminated the results (§11.4.119). The prior session ran three full `go test ./...` sweeps simultaneously on one host. Captured proof of contamination: `112× cannot assign requested address and 8× bind: address already in use`, with sibling sweep PIDs verified by `cwd/argv`. Test sweeps that bind ports are NOT parallelisable — partition by RESOURCE, not by repo. **Several reported failures are ambient-suspect, not proven code defects.** A serial, host-quiet re-run is REQUIRED before any of them are called real.

Correction 2 — docs/research/go-elder-plinius-v3 is NOT vendored. It has no nested `.git` and no LICENSE; the nested repos under `docs/research/` belong to SIBLING directories (Gandalf-Solutions, CL4R1T4S, LEAKHUB, AutoRedTeam, CLAUDE-CODE-SYSTEM-PROMPT). Reverting `gofmt` there was still CORRECT, but for a different reason: `go list ./...` returns **0** packages from it — it is archival research intake outside the root module graph that no `build/vet/test` gate can validate.

Correction 3 — G8 was a false-positive gate, not a documentation defect. Both items already carried complete, truthful Obsolete-Details. The gate asserted `Triple-check::canonical (constitution/Constitution.md:7687)` and the generator (`obsolete.go:187`) both say `Triple-check evidence::`. That markdown is machine-generated from the SQLite SSoT — hand-editing it would have broken **G11** (byte-identical `md` → `db` → `md` round-trip) and been reverted at next regeneration.

G7 (§11.4.83) — RESOLVED, 0 violations (audit re-verification, 2026-07-28T12:34Z)

G7 **now PASSES**. Independently re-run this audit: `bash scripts/verify_qa_evidence.sh --enforce --since 925169c98945ca0fee1e84dae53ad494e4897832` → **28 feature-shipping commits evaluated, 0 opt-outs, 0 violations**, RESULT: PASS. This supersedes the entire section below, which is PRESERVED as the historical record of how the 13-violation figure was reached and cleared — do not act on it as current state; every commit it lists as “still violating” (including all 3 security fixes 4727a9d0/9c876819/2ff55c31) now has `docs/qa/<run-id>/EVIDENCE.md` on disk and reports ok.

Historical section (superseded — preserved for provenance)

Prior text described the situation as **13** violations, not 24 (Class C was FIXED at that point; Class A/B were not yet):

Two stale figures were in circulation and both are superseded. The “24 violations” three-way split below was the original analysis; the 2026-07-27 sweep log records **18**; the freshest gate detail output records **13**. Provenance, so you can re-derive rather than trust:

Source	Count	Timestamp
Original three-way analysis (6+3+15)	24	2026-07-27 13:34Z log
qa-results/full_retest/verify_rules_20260727T184002Z.log RESULT line	18	2026-07-27 23:40
/tmp/g7-qa.out VIOL lines — freshest	13	2026-07-28 14:07

`/tmp/g7-qa.out` is the gate’s volatile detail file, **regenerated on every run and not durable**. Re-run `./scripts/verify-all-constitution-rules.sh` to regenerate it before relying on the 13.

- **Class C (6) — FIXED, confirmed.** These were FALSE violations (evidence existed; the gate could not match it — it required the COMMIT SUBJECT to contain the evidence directory basename, but run-ids are timestamped slugs, and flat `docs/qa/* .md` files were invisible to a directory-only enumeration). **All six now report ok:** 3e67fa1a, 54a76c3c, ca76e14b, 7210f373, 02e9e505, 1254e0a6 — verified individually in `/tmp/g7-qa.out`. This closes in-flight item #2 below.

- **Class B (3) — still violating**, all genuine non-features tripping the heuristic: d6c05f76 (README only), f9dcf6a6 (doc-comment only), b058c7c2 (const → var test-robustness, prod default unchanged).
- **Class A — 10 remain** (down from 15; fbfffd7d, 0e3bb747, 6efadd15, 225cdf77, aa6b20b4 now report ok because their EVIDENCE.md cites the commit SHA). Still violating, **including all 3 security fixes**: 4727a9d0 (CORS wildcard+credentials), 9c876819 (CSWSH /ws auth), 2ff55c31 (wire-facade 401). Full remaining list: eb233785 f8c38181 67c9a9bc f9dcf6a6 4727a9d0 9c876819 d6c05f76 2ff55c31 a21ad7ca 51c058b1 66f9c21e c9bad26a b058c7c2.

HARD CONSTRAINT: the [no-qa-evidence] opt-out lives in the COMMIT MESSAGE and every violating commit is already pushed. Retroactive application requires history rewrite — **forbidden absolutely** (§11.4.113 / CONST-043). The opt-out is available only going forward.

Class A is DECIDED (operator, 2026-07-28): clear G7 by **real retrospective QA runs, the 3 security commits first** — not by a documented baseline bump. The evidence must come from genuine runs performed now and committed under docs/qa/<run-id>/. Class C is already fixed, so the debt is no longer overstated.

Open defects — confirmed, NOT yet fixed

RESOLVED since this section was written (audit 2026-07-28T12:34Z, verify before re-opening): - §11.4.157 carrier lockstep gap — 5/6 carriers already reach §11.4.235 at HEAD; CRUSH.md's uncommitted working-tree edit already reaches it too. See "Live-state anchors". - ~~Constitution pin 6 commits behind~~ — now matches the checkout exactly (ce3331a1). See "Live-state anchors". - ~~G7 13 violations~~ — 0 violations, see the G7 section above.

Still open / unverified this audit:

- **IPv6 bracket-unawareness — TWO independent bugs, one class, neither a regression** (no shared helper; both non-bracket-aware since creation):
 - helix_code/internal/llm/response_err_round53_test.go:869 mustSplitHostPort re-joins the output of net.SplitHostPort (which strips IPv6 brackets by design) with plain concatenation → http://::1:PORT.
 - helix_agent/internal/vectordb/qdrant/client_mock_test.go:31-35 uses strings.Split(url, ":") on a bracketed authority → host == "[", and a discarded Sscanf error silently defaults the port to 80 → http://[:80.

- Trigger for both: `httptest.NewServer`'s IPv6 fallback when IPv4 loopback bind fails. **These tests are environment-flaky by construction.**
- **i18n key leak — LATENT, not live.** `tool_schema` ships a consumer-injected translator seam; **nothing anywhere calls `toolschema.SetTranslator`**, so `activeTr` stays `NoopTranslator` and `tr()` echoes the key. Bundle entry EXISTS and is correct (`active.en.yaml:50` → "Create git commit"). No production path currently reaches it, so no user sees it — but it arms on first use, and all 36 `tr()` sites include user-visible `ToolResult.Error` payloads. **Do NOT “fix” the test to expect the raw key** — that cements the unwired state and disarms the only cross-module check.
- **HXC-131 evidence path is dead** — `docs/qa/followup_fixes_20260712T085616Z/HXC131_evidence.md` never existed (that commit added only `HXC133_evidence.md`). Genuine evidence IS at `tracked scratch/discovery/fixes/HXC131_evidence.md`. Correct repair is a DB mutation on the SSoT (`workable-items obsolete-details --evidence ... + re-render`), not a doc edit.
- **`sdk/web/node_modules/flatted/golang/pkg/flatted/flatted.go`** — genuinely third-party vendored code sitting INSIDE the `gofmt` gate. Clean today; an `npm` bump reintroduces the tension.
- **`go vet ./...` covers only the root module** — 36 nested modules (ours included) are never vetted. Real coverage hole, own item.
- ~~`$11.4.157` carrier lockstep gap / Constitution pin 6 behind~~ — **both RESOLVED**, see the “RESOLVED since this section was written” note above this list. (Historical detail preserved: the gap was `constitution/Constitution.md` defining `$11.4.235` while carriers topped out at `$11.4.234`, compounded by the pin `731bf1d3` predating checkout `32d75788`. Both are fixed at current HEAD.)

NEW this revision (rev 7, 2026-07-28T17:37Z — found while re-deriving live state, not fixed in this pass, doc-only scope):

- **Disclosed RED bluff, unrepaired — `3fd55a4d`'s F1 regression guard.** `llm_generate_native_model_regression_test.go`'s `RED_MODE=1` branch replicates the pre-fix response construction locally and asserts on that replica, so it PASSES even on the already-fixed artifact — a RED that cannot fail is evidence-shaped, not evidence (`$11.4.1/$11.4.115`). The commit message discloses this itself and tracks a follow-up to rewrite the RED branch to drive the real handler against pre-fix behaviour (the same repair pattern applied to `debate_orchestrator`'s `protocol_convenience_form_red_test.go` in that submodule's `b0655d3`, per the dispatching brief — not independently re-verified by this pass). The `RED_MODE=0` GREEN half is genuine and load-bearing; only the RED half is a bluff.

- **Three `main_racefix_test.go` files remain authored-but-never-executed on this host** (`helix_code/applications/{desktop,harmony_os,aurora_os}/main_racefix_test.go`) — this host has no X11/GL dev headers so `go build` (without `-tags nogui`) fails for all three packages identically, and the `nogui` build tag excludes these files entirely (confirmed: they carry `//go:build !nogui`). **This gap is proven to hide real defects, not merely theoretical risk** — the `harmony_os` file was found, by `READING` alone, to panic deterministically on first execution (a nil-pointer dereference in `refreshData()` when `apiClient` is nil), and was fixed in `99ff7d8e` — still never actually run. The `desktop` and `aurora_os` files were independently traced (not executed) and judged safe under both `RED_MODE` settings; that judgement is unconfirmed by execution. Closing this gap needs either an X11-capable host or a from-scratch `nogui`-compatible rewrite (out of scope, no such rewrite exists).
 - **No `-count>=10` stability sweep exists for tests/scaling at its current `Trials=5` setting.** `c7484cc7` raised `MeasureTrialsPerStep` `3→5` for signal-to-noise (confirmed in source: `tests/scaling/scaling_harness.go:177`) with a handful of direct reruns pasted in its own commit message, but no durable `qa-results/` artefact records a `-count>=10` repetition. The dispatching brief for this task additionally reported a run that hung at `~0.3%` CPU and was killed without diagnosis; **this pass could not corroborate that specific detail from any log, process, or docs/qa artefact** — carried forward as reported, not independently confirmed, and should not be treated as resolved or as ruled out.
 - **`debate_orchestrator` has only ONE configured remote** (`origin → GitHub`), while every sibling submodule fans out to 4 (`github/gitlab/origin/upstream`). Flagged for the operator per the dispatching brief's framing — a decision point, not a defect this pass is authorised to fix.
 - **Independent review of `helix_agent`'s newest 2 commits is unconfirmed.** `129094b0` and `f451d342` landed after the reported 9-commit/3-round CLEAN GO ending at `0367570e`; no review artefact for these 2 was found by this pass. `f451d342` itself does textually remediate a Critical finding (see the Live-state anchors `helix_agent` row) — the fix reads as correct on inspection, but “reads as correct” is not the same as a completed independent review.
-

In flight at handoff — status re-derived 2026-07-28 (§11.4.147: a crashed agent is NOT complete)

Four background agents were dispatched at the 2026-07-27 handoff. Re-derived status:

1. Independent review (§11.4.142) of the llamacpp fix + its RED test — **NOT VERIFIED**. No review artefact located this revision. Treat as outstanding; re-dispatch. Re-checked rev 7 (2026-07-28T17:37Z): still no matching artefact found under docs/qa/; unchanged.
2. G7 Class-C matching-bug reconciliation + paired mutation — **LANDED, verified**. All six former Class-C commits now report ok in the gate detail output (see the G7 section). The paired mutation was **not** separately verified.
3. Extend the discarded-error guard to 4 sibling sites (internal/verifier/embedded_server.go:84, together/client.go:72, replicate/client.go:87, 129), TDD RED-first — **NOT VERIFIED**. Re-checked rev 7: internal/verifier/embedded_server.go:84 STILL discards the error (req, _ := http.NewRequestWithContext(...)), unaddressed. together/client_nilrequest_test.go and replicate/client_nilrequest_test.go DO exist, but both predate this remediation cycle (added in commit 56f7edf7, an old “Auto-commit”) — they were not verified to satisfy this specific item’s scope. Treat as still open.
4. Apply the prepared carrier elision patch (48 entries, 6 carriers) + verify §11.4.157 lockstep + regenerate exports — 5 of 6 carriers now RESOLVED (see Live-state anchors “Constitution ceiling vs carriers”, reconfirmed unchanged rev 7); CRUSH.md’s committed HEAD still stops at §11.4.234 while its own working-tree edit already reaches §11.4.235 — one commit away, still NOT LANDED.

Items 1 and 3 must be re-dispatched or re-verified — do not assume completion. Item 4 is nearly closed (one commit).

Uncommitted work in the tree (snapshot 2026-07-28 — VOLATILE, re-derive)

Not a backlog of unpushed commits — in-flight working-tree state. Account for every file before any `git add` (§11.4.84); never `git add -A`.

SUPERSEDED COUNT AGAIN (rev 7, 2026-07-28T17:37Z): 73 porcelain entries (46 untracked + 27 modified), not ~68. Re-derived fresh via `git status --porcelain | wc -l`. Continuing the same pattern rev 6 already described: mostly more new docs/qa/<run-id>/ evidence directories from the same ongoing G7-evidence stream, expected and not concerning. This task’s own scope

fence additionally confirms these ~73 entries (minus this doc pair) belong to OTHER active streams (helix_code/internal/llm/, helix_code/applications/, and others) — do not stage them, ever, and never `git add -A`. The itemised lists below (21 modified / 5 untracked) are the rev-5 SNAPSHOT, now doubly stale against the current 73; re-run `git status --porcelain` yourself rather than trusting this itemisation. Two items below are also stale: constitution (gitlink) is annotated “6-behind pin” below — that is RESOLVED (see Live-state anchors, pin now matches checkout); submodules/helix_agent (gitlink) is listed as modified — confirmed still true, and additionally the submodule’s own internal working tree is separately dirty (18 files this pass, see Live-state anchors), which this list does not capture (it only reflects the meta-repo’s view of the gitlink pointer, not the submodule’s internal state).

Modified (21, incl. this doc pair): .docs_chain/contexts/fixed.yaml, .docs_chain/contexts/issues.yaml, .superpowers/sdd/progress.md, RESUME.md, config/llmsverifier/config.yaml, constitution (gitlink — the 6-behind pin), docs/CONTINUATION.md, docs/SYSTEMD.md, docs/scripts/verify_qa_evidence.md, helix_code/config/replica-8081.yaml (operator decision 1), helix_code/internal/llm/response_err_round53_test.go, helix_code/internal/tools/shell/{executor,output}.go, helix_code/tests/e2e/challenges/{functional_validator, helixcode_server_client}.go, scripts/run-all-tests.sh, scripts/systemd/helixcode-server.service (operator decision 3 — secret env names), scripts/systemd/helixllm-gateway.service, scripts/tests/verify_qa_evidence_meta_test.sh, scripts/verify_qa_evidence.sh, submodules/helix_agent (gitlink).

Untracked (5): docs/qa/phase1_fullhttp_e2e_20260728T091257Z/ and docs/qa/phase1_fullhttp_e2e_20260728T093918Z/ — 5 captured HTTP evidence files each (401-no-auth, OpenAI chat-completions 200, Anthropic messages 200, OpenAI tool-calls 200, Anthropic tool-use 200); helix_code/internal/tools/shell/stream_race_guard_test.go; helix_code/tests/e2e/challenges/helixcode_server_client_hostport_test.go; scripts/lib/.

Traps that already cost time — do not re-learn these

- A “package ok” may be ambient, not repair. internal/llm reported ok after the crash fix — including 3 round-53 tests that had failed. Those pass because httptest bound IPv4 that run, NOT because the IPv6 defect was fixed. It is unfixed. Green here is environment.

- **The legacy summary generators DESTROY data.** `scripts/generate_{issues,fixed}_summary.sh` read the TEXT trackers; committed summaries are SQLite-derived (§11.4.93/.95). Running them rewrites 344 items down to 188 — 156 items of tracked state destroyed.
- **make test can never pass on a headless host** — runs `go test ./...` with no `-tags=nogui`.
- **A “completed / exit 0” notification describes the WRAPPER, not the work.** Verify the artefact — `mtimes`, `pgrep`, pasted output.
- **replace_all is unsafe when a NEW helper’s body contains the pattern being replaced** — rewrites the helper into infinite self-recursion.
- **PDF export silently drops wide tables’ rightmost column** (§11.4.168) while reporting `rendered=2 failed=0`. Keep load-bearing evidence out of wide tables.
- **A §11.4.115 RED test that asserts against a local replica of the old behaviour, instead of driving the real production code path, is a bluff gate — it can never fail, so a PASS proves nothing.** Found (at least) twice this session in this exact shape: `3fd55a4d`’s F1 guard replicates the pre-fix `generateLLM` response construction locally rather than calling the real handler; per the dispatching brief (not re-verified by this pass) `debate_orchestrator’s protocol_convenience_form_red_test.go` had the identical defect, repaired in `b0655d3`. When authoring or reviewing any RED/GREEN polarity test, check specifically whether the RED branch calls into the actual function/handler under test.
- **A logo-fix-class subagent commit can sweep an unrelated file’s staged state into an unintended commit if git add is not scoped to explicit paths** — the reason §11.4.84 forbids `git add -A/git add .` in a shared checkout with concurrent streams. This session’s scope fence exists specifically because of this risk class; when in doubt about what a broad `git add` would sweep in, stage explicit paths only and re-check `git status` immediately before `git commit`.
- **git ls-files '*.go' | xargs gofmt -w reaches vendored AND archival trees.** Always exclude `docs/research/` (and check `node_modules`) before formatting.
- **podman pod rm -f pod_helix_agent KILLS helixagent.service.** Its container adapter sees the pod vanish and shuts down gracefully (`"Shutting down container adapter..."`, `status=0/SUCCESS`) — it looks like a clean exit, not a casualty. Cost this session: `chaos/api + chaos/auth` regressed ok → connection refused on `:7061`, and a whole 30-min sweep ran against a dead server. Stop the service first, or don’t touch its pod.
- **pgrep -f 'go test' matches YOUR OWN shell** — the loop’s command line contains the literal string. It killed my own bash mid-command (exit 144). Always exclude `$$/$PPID` and verify by `cwd` (§11.4.174 / §11.4.201 carrier footgun; §12.12 names this exact class).

- **:0 failing with EADDRINUSE is the ephemeral-exhaustion tell.** Binding port 0 asks the kernel for *any* free port; it can only fail when the range is genuinely exhausted. That single line is the fastest way to distinguish host contention from a real allocator bug.
 - **The test stack and the platform stack use DIFFERENT ports.** Precondition demands 15432/16379/18081 (docker-compose.test.yml); the live platform uses 5433/6380 (helixcode-infra-*). Seeing 5432/6379 listening proves nothing about test readiness — those belong to another project.
-

Sweep commands (exact) — RUN SERIALY, ONE AT A TIME

```
# Host must be quiet. Three of these in parallel exhausts IPv4
ephemeral
# ports, forces httptest onto its IPv6 fallback, and produces
failures that
# are contention artefacts rather than code defects (§11.4.119).
cd helix_code                && go test -tags=nogui -count=1 ./...
cd submodules/helix_agent    && go test -count=1 ./...
cd submodules/helix_qa       && go test -count=1 ./...
./scripts/verify-all-constitution-rules.sh
./scripts/verify-governance-cascade.sh
```

-tags=nogui is REQUIRED for the inner module (headless host, no X11/GL dev headers). -count=1 is REQUIRED — an earlier run reported 250 ok of which 249 were (cached).

Binding constraints (restated — §11.4.131(C))

- **Anti-bluff §11.4** — every PASS needs captured runtime evidence; metadata-only / config-only / absence-of-error / grep-without-runtime PASS are critical defects.
- **§11.4.113 — force-push is absolutely forbidden**, no exception, no operator override. Integrate by merging onto latest main, push fast-forward only.
- **§11.4.151** — tags carry the helix-code- prefix, identical across main repo and every owned submodule in one release.
- **§11.4.142** — every change needs INDEPENDENT review before commit/build; author self-verification (§11.4.92) precedes but never satisfies it.
- **§11.4.84** — account for every modified file before git add; unaccounted entries → ABORT. Never git add -A here.
- **§11.4.119** — one owner per exclusive resource; partition parallel work by RESOURCE, not by repository.

- **§11.4.185** — manual QA-team confirmation is the final sufficiency gate; not self-certifiable.